

Blood Donor Registry and Availability Search

Full Stack Web Application for Efficient Blood Donor Management

Varshini V — Artificial Intelligence and Data Science

Prince Dr. K. Vasudevan College of Engineering and Technology

Blood Donor Registry

 **Dashboard**

 **Donors**

 **Add Donor**

 **Donation History**

 **Eligibility**



Problem Statement

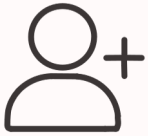
Hospitals often maintain donor information manually. In emergencies, locating eligible donors by blood group and location is time-consuming because records are scattered, outdated, or lack donation history.

- **Time-sensitive**
Manual lookup delays donation outreach during critical windows.
- **Stale records**
Outdated donor details lead to false matches and wasted effort.
- **Eligibility verification**
Donation eligibility (intervals, history) is hard to verify manually.

❑ Objective: Build a searchable registry that stores history and auto-checks eligibility by blood group, area, and donation interval.

Proposed Solution

A full-stack web application that centralizes donor records, preserves donation history, and returns eligible donors quickly based on configurable rules.



Register Donors

Collect personal details, blood group, and initial donation data.



Search & Filter

Query by blood group, area, and eligibility window in seconds.



Donation History

Store all donation events—no overwrites — enabling accurate eligibility checks.



Dashboard

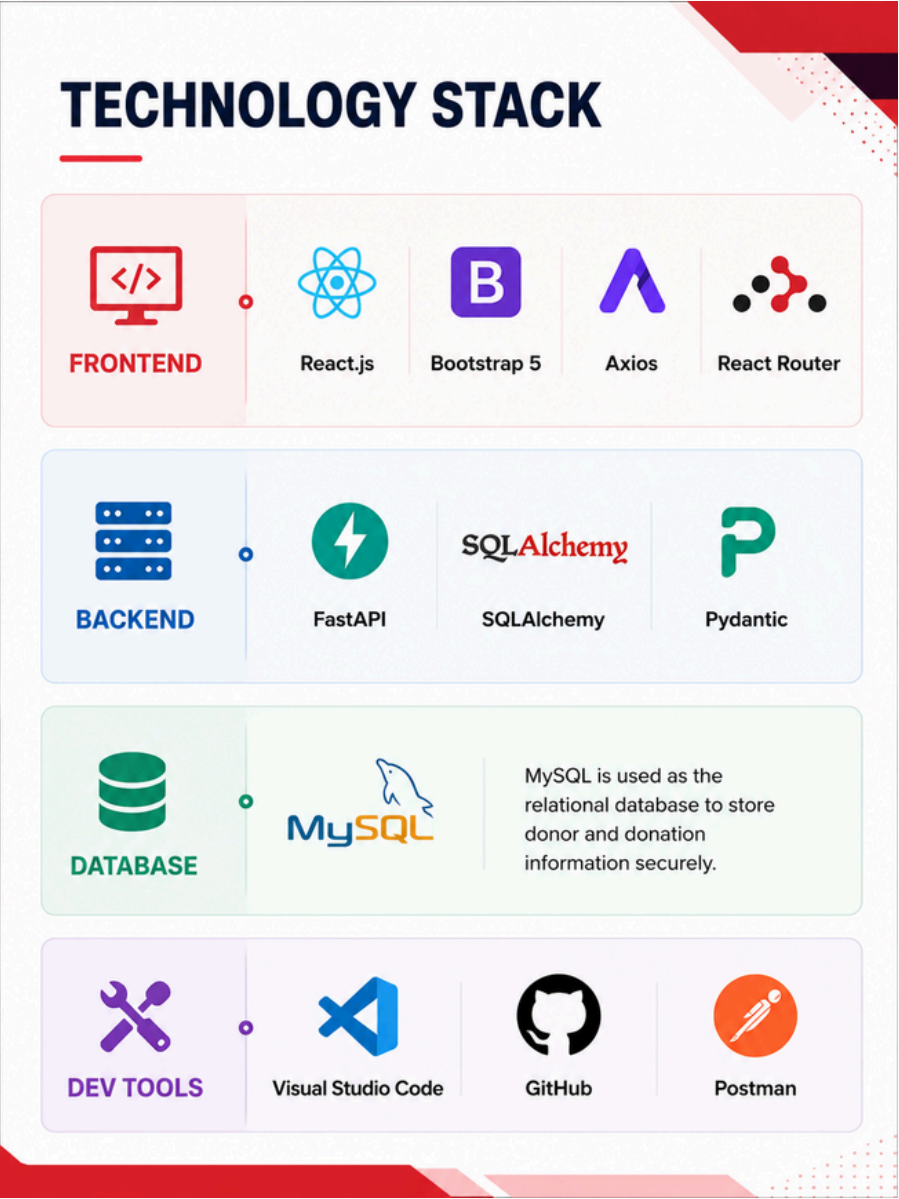
At-a-glance statistics: donors by group, recent activity, and availability.



Manage Records

Edit or remove records with role-based controls.

Technology Stack



Frontend

React.js • Bootstrap 5 • Axios • React Router



Backend

FastAPI • SQLAlchemy • Pydantic



Database

MySQL — relational data model for donors and donation history

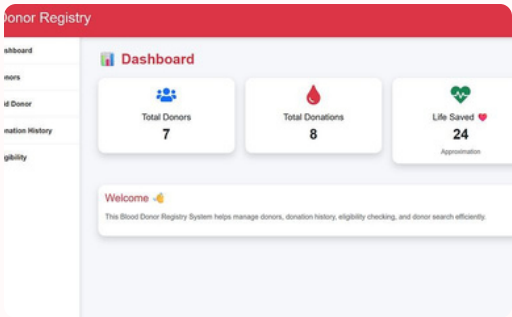


Dev Tools

Visual Studio Code • GitHub • Postman

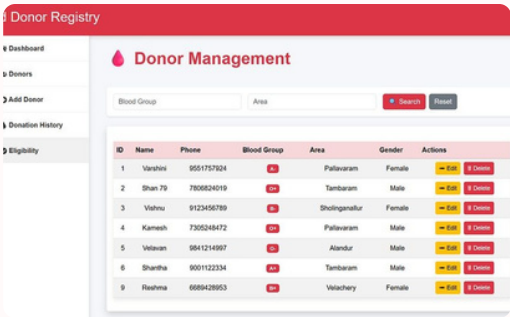
System Features

Key screens and workflows



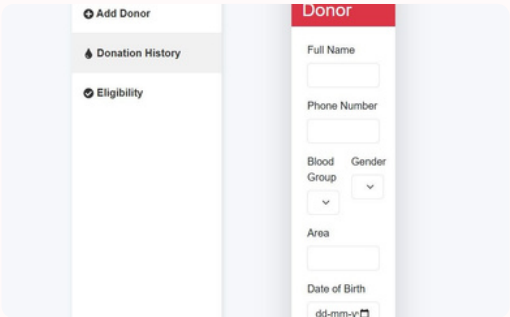
Dashboard

Overview: donor counts, eligibility alerts, recent donations.



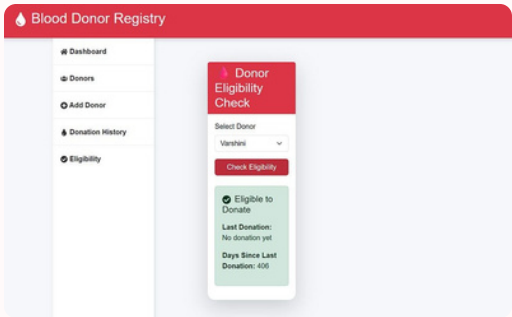
Donor Management

List, edit, delete donors; quick filters by blood group and area.



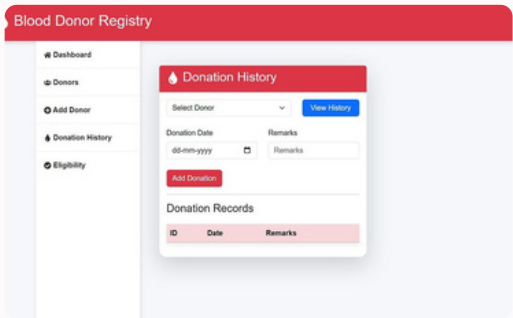
Add Donor

Structured registration form with validation and consent checkbox.



Eligibility Checker

Automated status with calculated days since last donation.



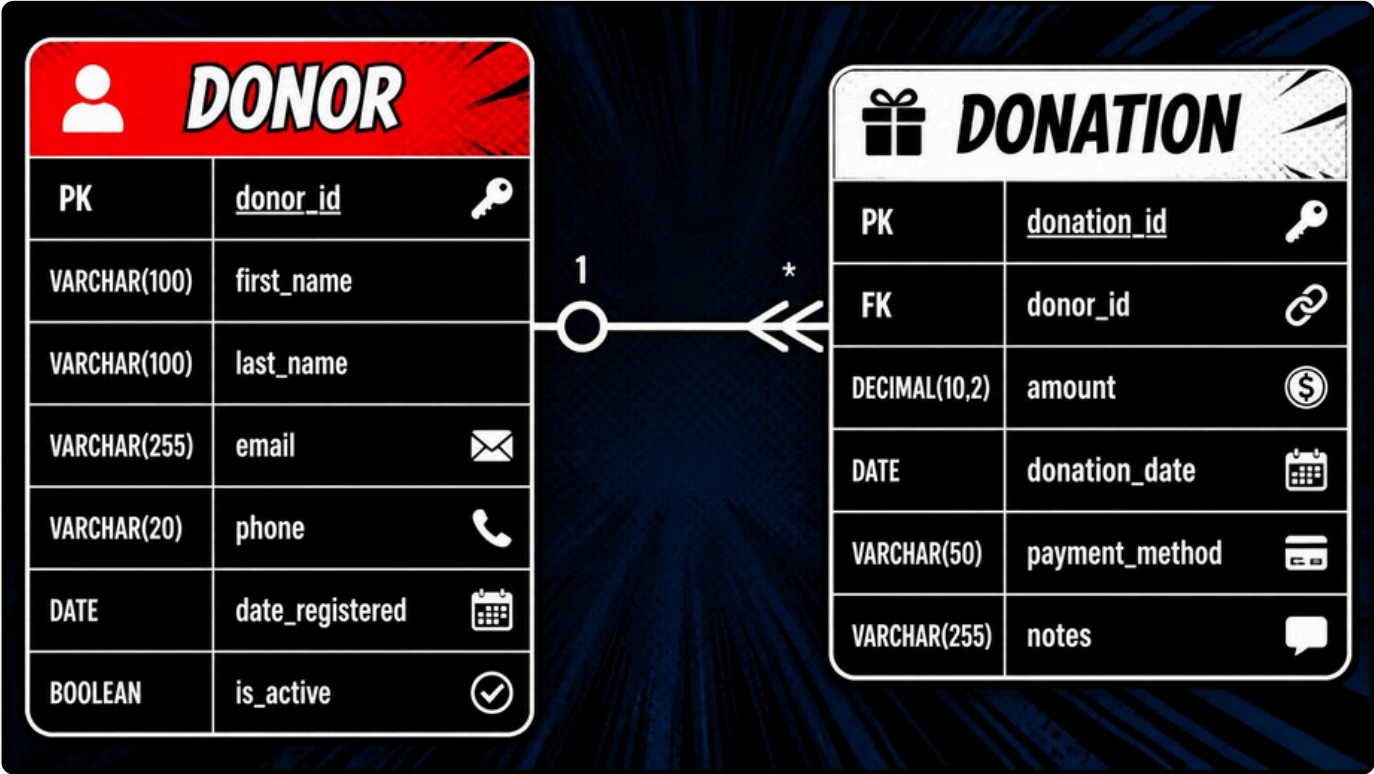
Donation History

Chronological records maintained per donor for eligibility checks.

Database Design

Core entities and attributes:

The relationship: One Donor → Many Donations.
The Donation table preserves history rather than overwriting last donation dates.



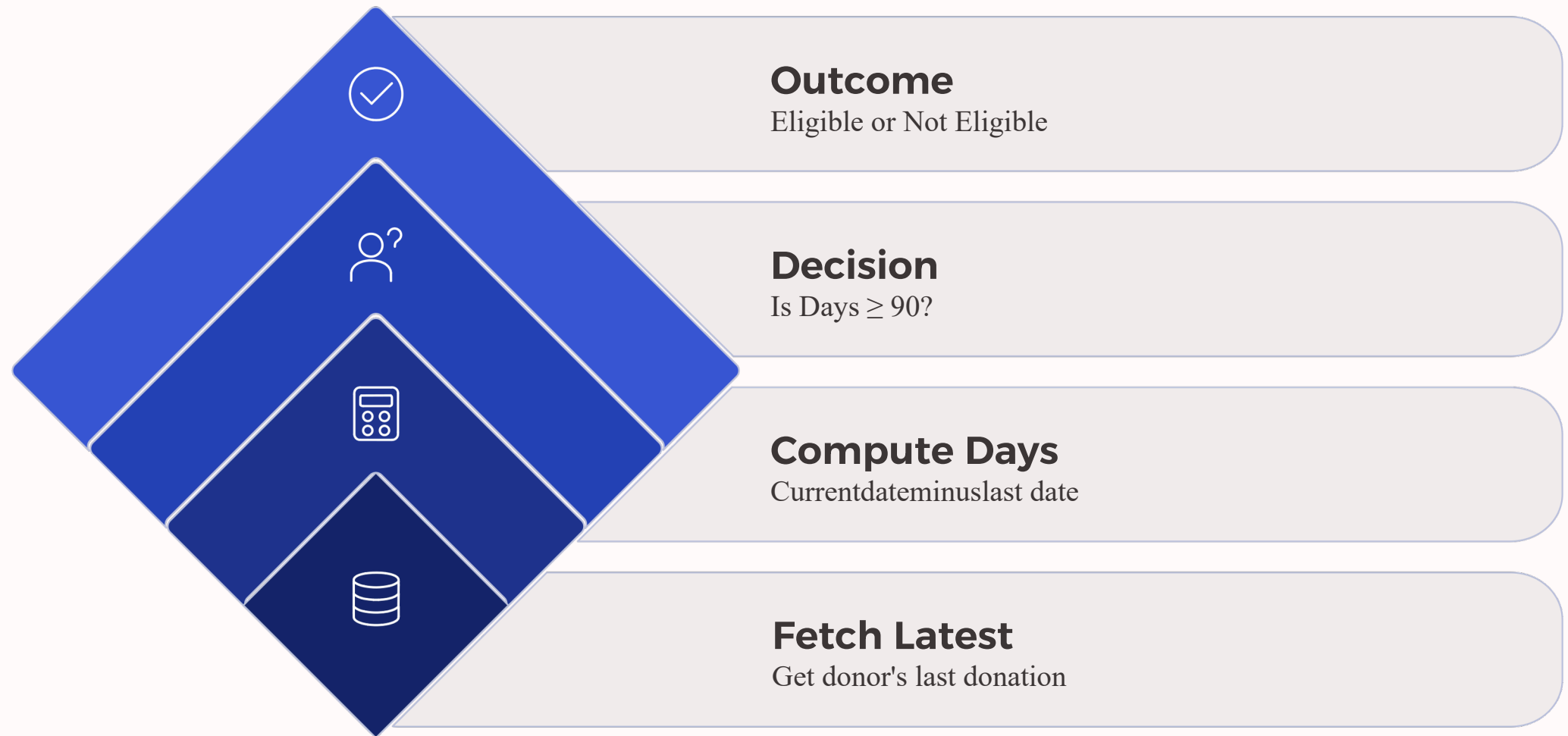
Donor

id, name, phone, blood_group, area, gender, dob

Donation

id, donor_id, donation_date, remarks

Donor Eligibility Calculation



Eligibility is evaluated server-side to ensure consistent results. Rule: if Days Since Last Donation ≥ 90 → donor is eligible prevents contacting ineligible donors and speeds hospital outreach.

1 Step 1

Fetch latest donation_date from Donation table.

2 Step 2

Compute difference with current date.

3 Step 3

Return eligibility flag and include in search results.

Conclusion & Future Improvements

TheBloodDonor Registry simplifies donor management, preserves donation history, and enables rapid identification of eligible donors in emergencies—reducing manual effort and response time.



SMS Notifications

Automated outreach to eligible donors.



Email Reminders

Scheduled reminders for upcoming eligibility.



GPS Nearby Search

Locate donors within a dynamic radius for faster response.



Auth & Roles

Implement authentication and role-based access control.



Blood Inventory

Integrate inventory data to match supply with demand.

Thank You